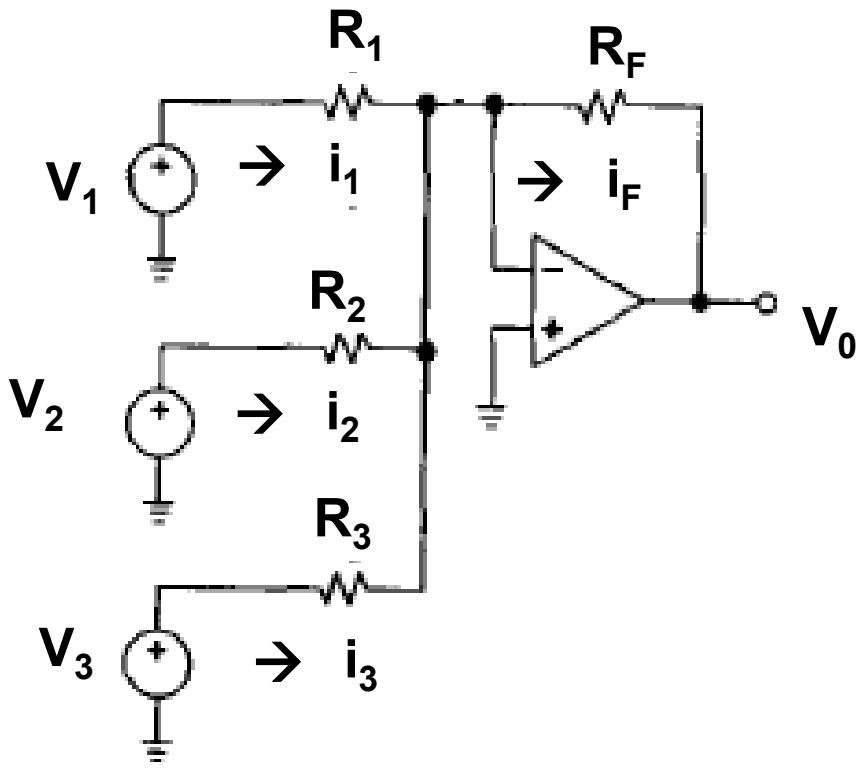
OP AMP

Summing Amplifier



$$i_1 + i_2 + i_3 = i_F$$

$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = -\frac{v_O}{R_F}$$

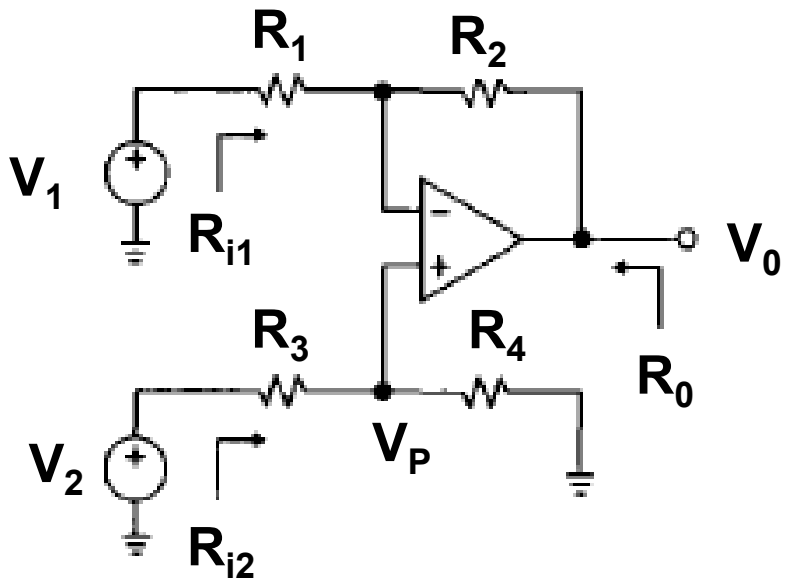
$$v_O = -\left(\frac{R_F}{R_1}v_1 + \frac{R_F}{R_2}v_2 + \frac{R_F}{R_3}v_3\right)$$

Using standard 5% resistances, design a circuit such that $v_O = -2(3v_1 + 4v_2 + 2v_3)$

$\frac{R_F}{R_1} = 6$ $\quad \frac{R_F}{R_2} = 8$ $\quad \frac{R_F}{R_3} \doteq 4$

If $R_F = 20k\Omega$,
$R_1 = 3.33k\Omega$
$R_2 = 2.50k\Omega$
$R_3 = 5.00k\Omega$

Difference Amplifier



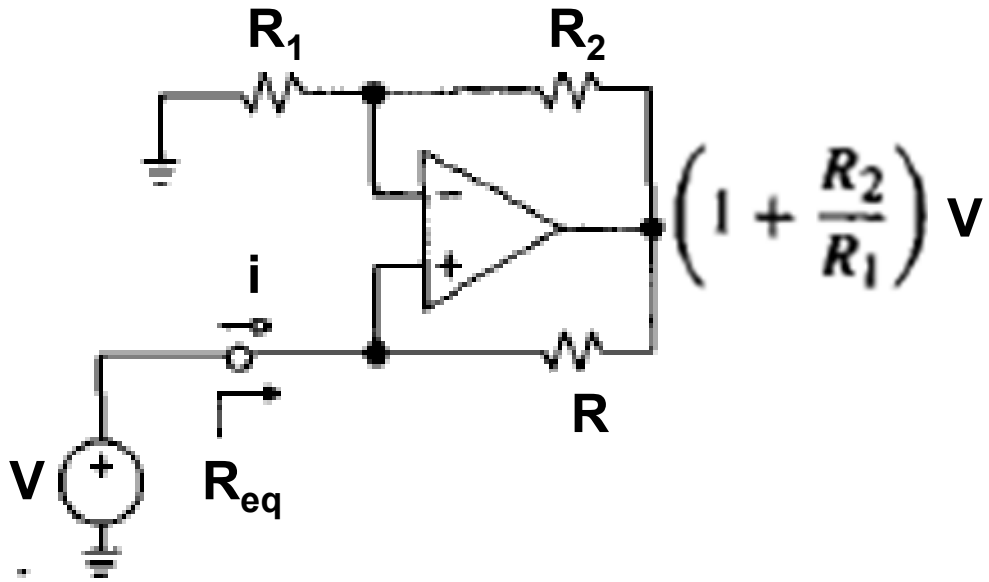
$$v_{O1} = -(R_2/R_1)v_1 \text{ and } R_{i1} = R_1$$

$$v_{O2} = (1 + R_2/R_1)v_P = (1 + R_2/R_1) \times [R_4/(R_3 + R_4)]v_2$$

$$R_{i2} = R_3 + R_4.$$

$$v_O = \frac{R_2}{R_1}\left(\frac{1 + R_1/R_2}{1 + R_3/R_4}v_2 - v_1\right)$$

When $\frac{R_3}{R_4} = \frac{R_1}{R_2}$ $\qquad v_O = \frac{R_2}{R_1}(v_2 - v_1)$



$$i = [v - (1 + R_2/R_1)v]/R$$

$$= -R_2 v/(R_1 R)$$

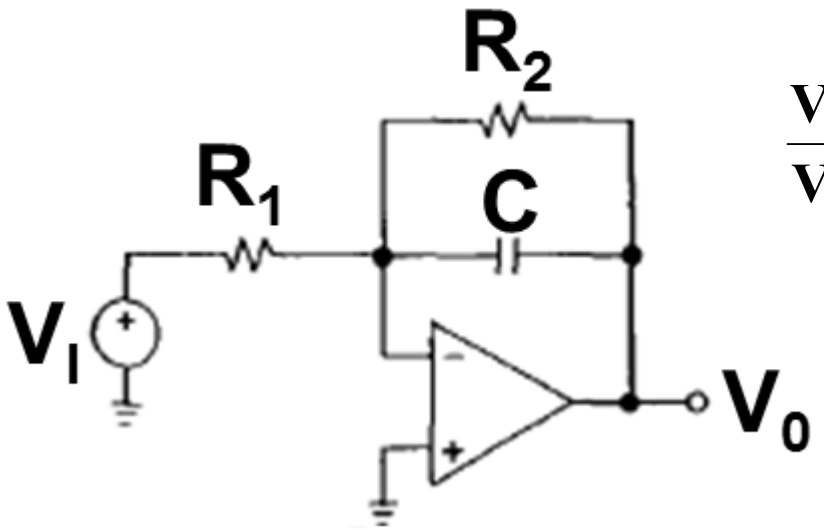
$$R_{eq} = v/i = -\frac{R_1}{R_2}R$$

the circuit simulates a *negative resistance*.

Negative resistances can be used to neutralize unwanted ordinary resistances, as in the design of current sources, or to control pole location, as in the design of active filters and oscillators.

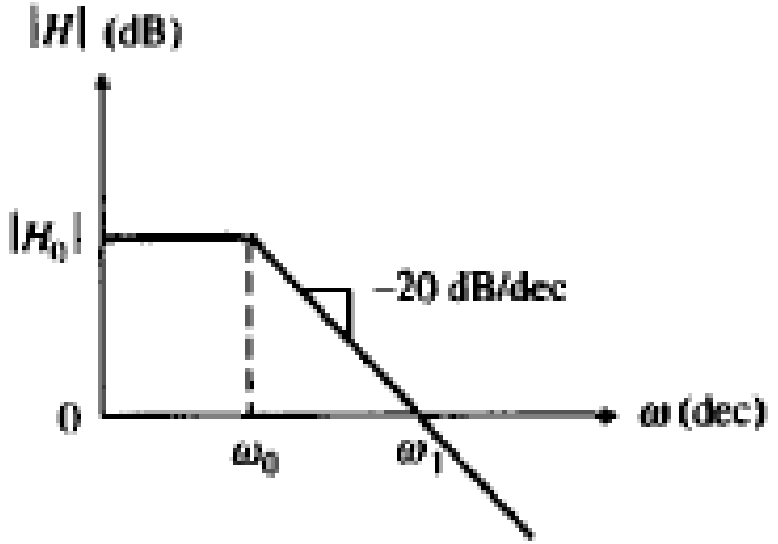
Active Filters

Low-Pass Filter with Gain

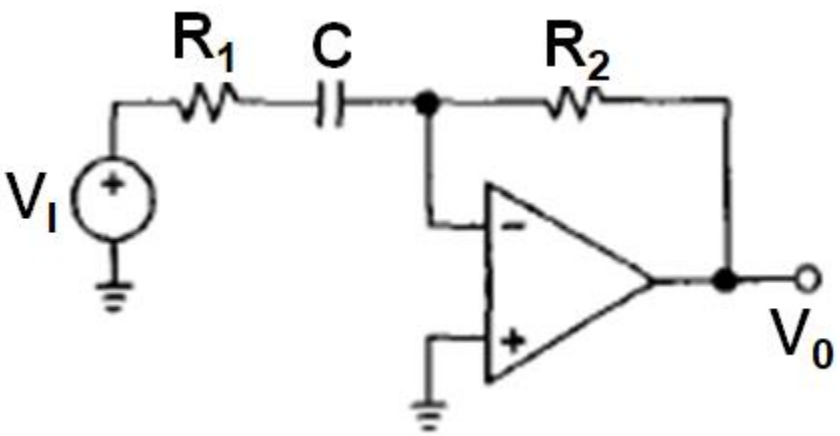


$$\frac{V_0}{V_I} = -\frac{\frac{R_2}{1+sCR_2}}{R_1} \Rightarrow H(s) = -\frac{R_2}{R_1}\frac{1}{1+sCR_2}$$

$$\Rightarrow H(j\omega) = H_0 \frac{1}{1+j\omega/\omega_0} \qquad H_0 = -\frac{R_2}{R_1} \qquad \omega_0 = \frac{1}{CR_2}$$

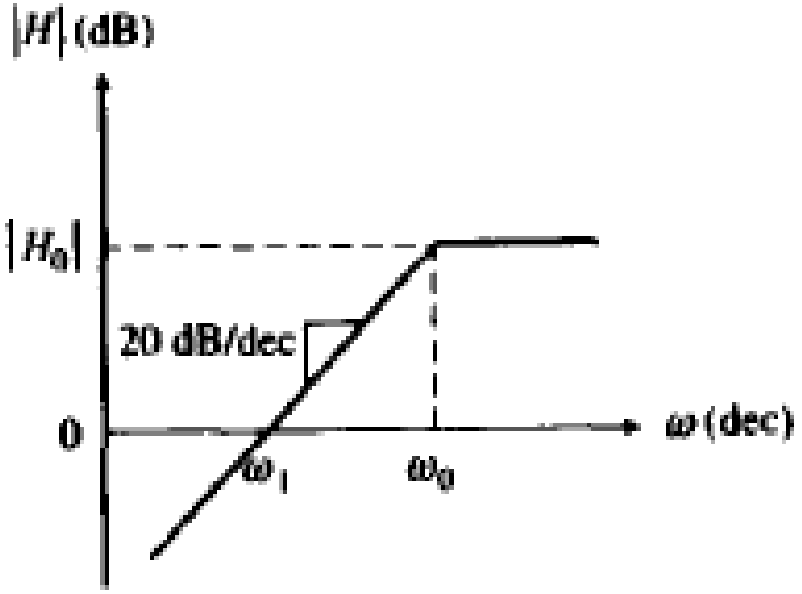


High-Pass Filter with Gain



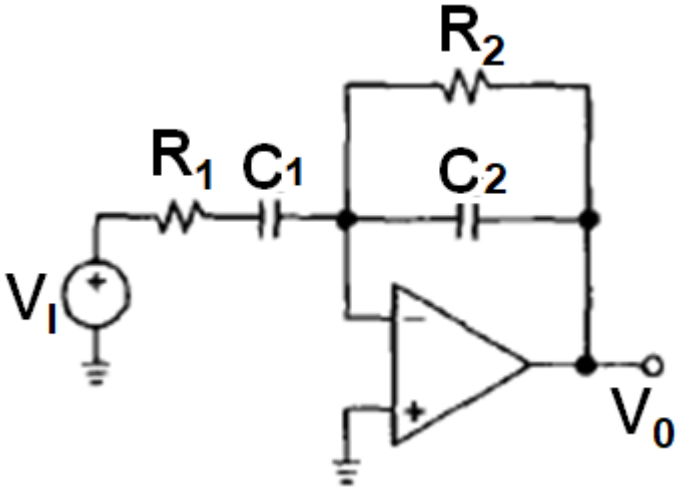
$$\frac{V_0}{V_I} = -\frac{R_2}{R_1 + 1/sC} = -\frac{sCR_2}{1+sCR_1} = -\frac{R_2}{R_1}\frac{sCR_1}{1+sCR_1}$$

$$H(j\omega) = H_0 \frac{j\omega/\omega_0}{1+j\omega/\omega_0} \qquad H_0 = -\frac{R_2}{R_1} \qquad \omega_0 = \frac{1}{R_1 C}$$



H_0 is High frequency gain
ω_0 is 3dB frequency

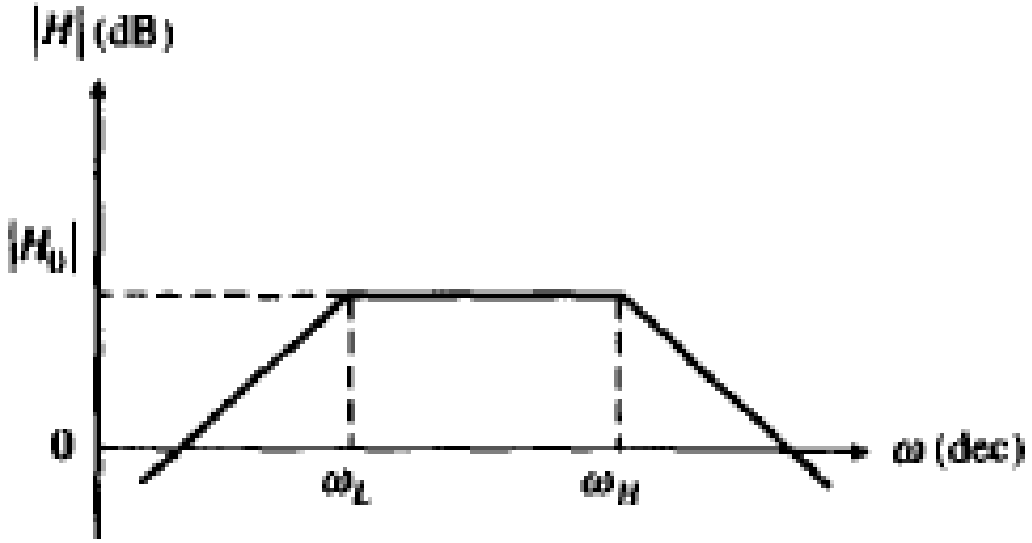
Wideband Band-Pass Filter



$$\frac{V_0}{V_I} = -\frac{R_2}{1+sC_2R_2} \Big/ \frac{1+sC_1R_1}{sC_1} = -\frac{R_2}{R_1}\frac{1}{1+sC_2R_2}\frac{sC_1R_1}{1+sCR_1}$$

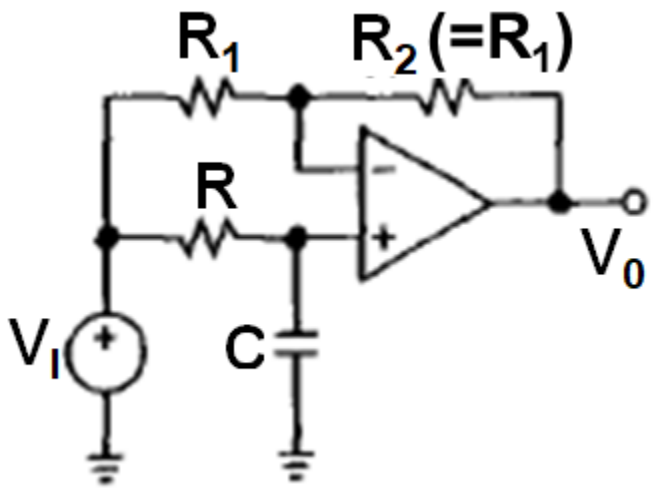
$$H(j\omega) = H_0 \frac{j\omega/\omega_L}{(1+j\omega/\omega_L)(1+j\omega/\omega_H)} \qquad H_0 = -\frac{R_2}{R_1} \qquad \omega_L = \frac{1}{R_1C_1}$$

$$\omega_H = \frac{1}{R_2C_2}$$



H_0 is mid-frequency gain
ω_L is lower 3dB frequency
ω_H is higher 3dB frequency

Phase Shifters



$$H(s) = 2\frac{1}{1+sCR} - 1 = \frac{1-sCR}{1+sCR}$$

$$H(j\omega) = \frac{1-j\omega/\omega_0}{1+j\omega/\omega_0} = 1\angle -2\tan^{-1}(\omega/\omega_0)$$

From $|H(j\omega)|$ point, it is a all pass filter

From $\angle H(j\omega)$ point it is a phase shifter

LPF $$H(j\omega) = H_0 \frac{1}{1 + j\omega/\omega_0}$$

HPF $$H(j\omega) = H_0 \frac{j\omega/\omega_0}{1 + j\omega/\omega_0}$$

APF $$H(j\omega) = \frac{1 - j\omega/\omega_0}{1 + j\omega/\omega_0}$$

Numerator decides type of filter

If numerator N(jω) =1 → LPF

If numerator N(jω) =jω/ω_0 → HPF

If numerator N(jω) =1-(jω/ω_0) → APF

H_0 is acting as scaling factor
Moves response up or down